



ECE 730
Control of Adjustable Speed Drives

Lab 4
**Mechanical sensor failure and
sensorless control of PMSM**

Due date: Apr. 13, 2025 (before 11:59PM)

Please upload report on Avenue to Learn:
Assessments> Assignments> Lab 4

Late submissions won't be marked.

Objective

- Simulating a mechanical sensor fault, developing a fault detection algorithm and a mechanical sensorless control of a three-phase Permanent-Magnet Synchronous Machine using MATLAB-Simulink.

Note- You are supposed to be familiar with MATLAB-Simulink and are able to run your simulation either remotely or on your computer.

Procedure

Figure 1 shows the block diagram of the current control loop in the framework of vector control of sinusoidal Permanent-Magnet synchronous machines (PMSM).

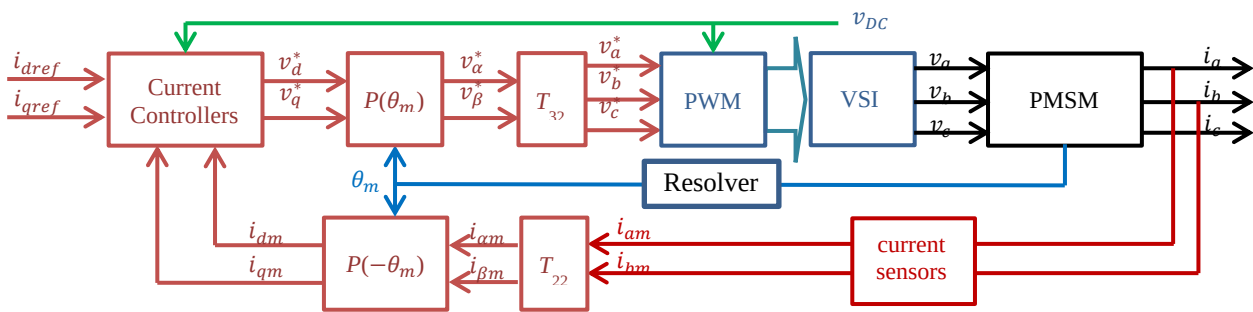


Figure 1- Vector control of PMSM: current control loop.

In this figure, the block PMSM has been already developed in Lab 1. Current measurements are supposed perfect and transformation matrices in the controller (blocks in red) are the same as those already used in the modeling of PMSM in Lab 1. The stator dq-currents control has been realized in Lab 2.

I. Position sensor (resolver)

The position sensor is a resolver. The concept is described in the Appendix. Main resolver faults are:

- Supply failure.
- Connector open-circuit.
- Disconnection from the shaft.
- Unbalanced amplification and filtering circuits.
- Measurement offset.

A Simulink program is available in the course website on Avenue. This program includes the resolver model. It includes a functional model of the resolver, a Resolver to Digital (R/D) converter (examples: AD2S1210 or AU6805) and a fault generation logic. The R/D contains a PLL which is implemented to extract the rotor electrical position and speed and to provide them to the control board.

- 1) Design the PLL to meet the following requirements:
 - Closed-loop bandwidth = 30 Hz
 - Phase margin > 45°
 - Gain margin > 6 dB
- 2) Run the simulation and observe the effect of the resolver on the current control performance in terms of set-point tracking, response time, overshoot and decoupling control.
- 3) What happens to the current control performance if the closed-loop bandwidth of the PLL is set to 300 Hz?
- 4) Keep the best design for PLL and study the effect of the following faults on the stator dq-currents:
 - Supply failure.
 - Unbalanced circuits, *e.g.*, 5%.
 - Measurement offset, *e.g.*, 10%.
- 5) Propose and implement a fault-detection algorithm to detect the resolver fault. Test the fault-detection algorithm and its sensitivity to the above faults.

II. Sensorless control of PMSM

The above fault signal is sent to the control board. It will be used to switch between the actual position measurement and the estimated position. This latter is provided by a sensorless control to be implemented in the control board.

- 1) Select the “Proportional Sensorless Control” (see lecture notes) and implement it in Simulink.
- 2) Consider the resolver’s supply is off (no rotor position measurement). Run the simulation for $b = 2$ and observe the effect of the sensorless control on the current control performance in terms of set-point tracking, response time, overshoot and decoupling control.

Note: It is recommended to filter the (estimated) speed (ω_s) for any other use in the control program except the calculation of the estimated position (9). The bandwidth of the low-pass filter should be set to allow only mechanical frequencies to pass through.
- 3) To study the effect of the proportional gain b , consider two cases:
 - a. $b = 2$
 - b. $b = 0.5$
 - c. For each case, run the simulation and study the convergence of the sensorless control for several initial rotor positions ranging from -180° to $+180^\circ$.
- 4) If you noticed any convergence issue in Question II.3, describe your results (you can refer to lecture notes). Is there any solution to avoid wrong convergence? Explain.
- 5) Now, consider the resolver’s supply is on (rotor position measured). Run the simulation and suddenly turn off the resolver’s supply manually around $t = 0.25$ s. What happens to the stator currents? Why?

Appendix

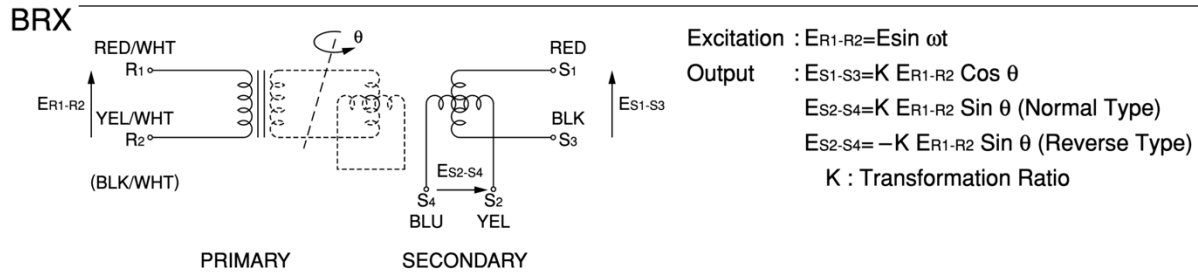
PRINCIPLE

Resolver is a rotary transformer, which outputs AC voltage in accordance with angular position of the shaft. There are two types of resolvers, BRX and BRT, having different types of winding.

BRX resolver is excited by AC voltage to the rotor winding, and outputs from the stator windings sine and cosine voltages proportion to the rotation angle θ .

BRT resolver is excited by sine and cosine voltages to the stator windings, and outputs from the rotor winding a sine voltage phase-shifted in proportion to θ .

The difference is illustrated as follows.



Reference: Tamagawa Resolvers (resolverssmartsyn.pdf)